EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 15

Basic Maths and Calculus

Introduction to Integration

By - Excellence Programme

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Topics to be covered

01 5-minute differentiation recap

02 Integration as reverse differentiation

03 Indefinite integral and +C

04 Power-rule reverse

05 Standard integrals table

06 Easy indefinite-integral practice

07 Physics link: a → v → x by integrating

What differentiation gave us

$d(x^n)/dx = n\,x^{n-1}$

$d(\sin x)/dx = \cos x$ · $d(\cos x)/dx = -\sin x$

$d(e^x)/dx = e^x$ · $d(\text{constant})/dx = 0$

$v = dx/dt$ · $a = dv/dt$

Quick check: differentiate

Differentiate $y = 3x^4 - 2x + 5$.

Solution: Quick check: differentiate

Step 1: $d(3x^4)/dx = 12x^3$

Step 2: $d(-2x)/dx = -2$ and $d(5)/dx = 0$

$dy/dx = 12x^3 - 2$.

Rough work / extra work

Integration: reverse of differentiation

Differentiation finds a rate from a quantity.

Integration finds the original quantity from a rate.

If $\frac{dy}{dx} = f(x)$ then $y = \int f(x)\,dx$

We read $\int f(x)\,dx$ as “the integral of f with respect to x”.

The constant of integration

Many functions share the same derivative, because a constant disappears when we differentiate.

$$\int f(x)\,dx = F(x) + C$$

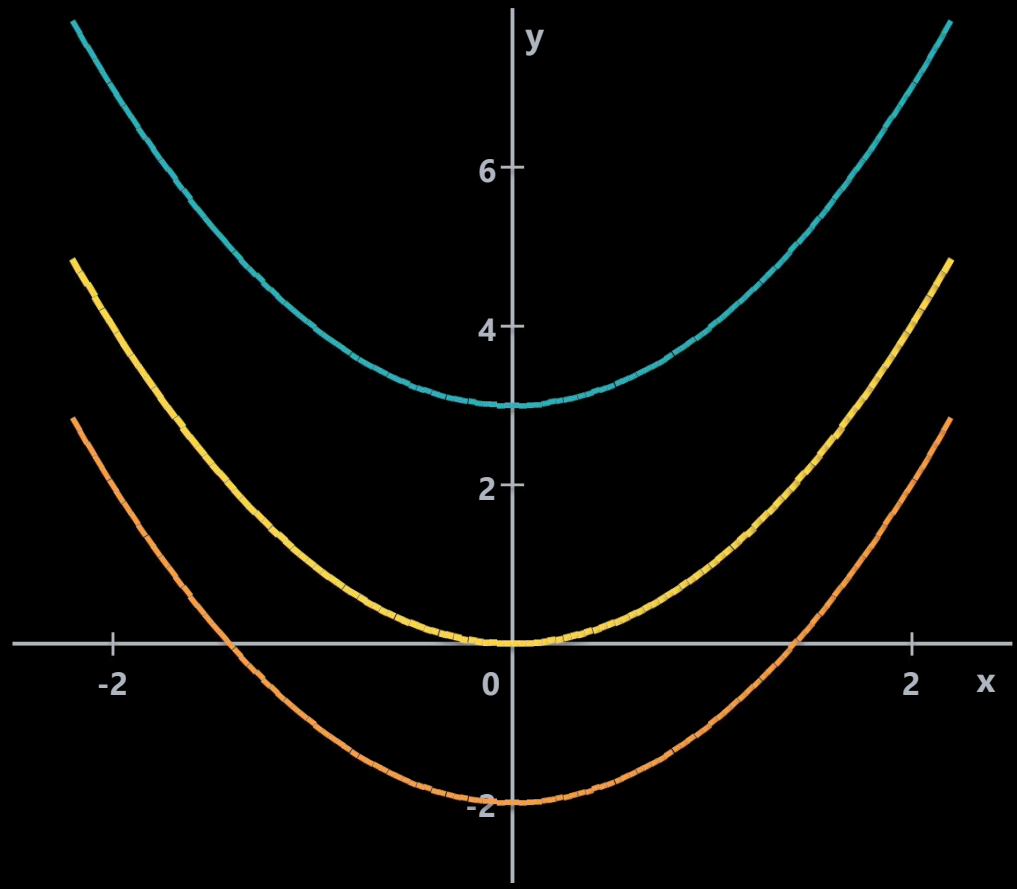
F(x) is any antiderivative of f. C is an arbitrary constant.

Never drop +C in an indefinite integral.

Same slope, different heights

$y = x^2$, $y = x^2 + 3$ and $y = x^2 - 2$ all have slope $2x$.

$d(x^2 + C)/dx = 2x$ for every C



same shape, shifted up/down

Power-rule reverse

Differentiation lowers the power by 1 and multiplies by the old power.

Integration raises the power by 1 and divides by the new power.

$$\int x^n \, dx = x^{n+1}/(n+1) + C \qquad (n \neq -1)$$

This formula fails when n = −1. That special case is ln|x| + C (later).

Integrate a power

Find $\int x^3\,dx$.

Solution: Integrate a power

Step 1: Raise power: $x^3 \to x^4$

Step 2: Divide by new power: $x^4 / 4$

$\int x^3\,dx = x^4/4 + C.$

Rough work / extra work

Integrate with a coefficient

Find $\int 5x^2\,dx$.

Solution: Integrate with a coefficient

Step 1: Constants come outside: $5 \int x^2 \, dx$

Step 2: $\int x^2 \, dx = x^3/3$

$\int 5x^2 \, dx = 5x^3/3 + C.$

Polynomial term by term

Find $\int (4x^3 - 6x + 2)\,dx$.

Solution: Polynomial term by term

Step 1: Integrate each term separately

Step 2: $\int 4x^3\,dx = 4 \cdot x^4/4 = x^4$

Step 3: $\int -6x\,dx = -6 \cdot x^2/2 = -3x^2$

Step 4: $\int 2\,dx = 2x$

$x^4 - 3x^2 + 2x + C$.

Rough work / extra work

Standard indefinite integrals

Function f(x)	∫ f(x) dx
x^n $(n \neq -1)$	$x^{n+1}/(n+1) + C$
1 (constant)	$x + C$
sin x	$-\cos x + C$
cos x	$\sin x + C$
e^x	$e^x + C$
1/x $(x \neq 0)$	$\ln\|x\| + C$

Check by differentiating

The safest check: differentiate your answer. You must recover the integrand.

$$\frac{d}{dx}\left[\int f(x)\,dx\right] = f(x)$$

If the derivative does not match f(x), the integral is wrong.

Always check +C disappears when you differentiate — that is correct.

Check an antiderivative

Is $F(x) = x^3/3 + 7$ an antiderivative of $f(x) = x^2$?

Solution: Check an antiderivative

Step 1: Differentiate F: $dF/dx = x^2$

Step 2: This equals $f(x) = x^2$

Yes. (Any constant in place of 7 also works.)

Integrate sine

Find $\int \sin x \, dx$.

Solution: Integrate sine

Step 1: From the table: antiderivative of sin x is −cos x

$$\int \sin x \, dx = -\cos x + C.$$

Rough work / extra work

Integrate cosine

Find $\int \cos x \, dx$.

Solution: Integrate cosine

$$\int \cos x \, dx = \sin x + C.$$

Integrate exponential

Find $\int e^x \, dx$.

Solution: Integrate exponential

Step 1: e^x is its own derivative and its own antiderivative

$$\int e^x \, dx = e^x + C.$$

Rough work / extra work

Mixed easy integral

Find $\int (2\cos x - 3\sin x)\,dx$.

Solution: Mixed easy integral

Step 1: $\int 2\cos x\,dx = 2\sin x$

Step 2: $\int -3\sin x\,dx = -3(-\cos x) = 3\cos x$

$2\sin x + 3\cos x + C$.

Simple reciprocal

Find $\int (1/x)\,dx$ for $x > 0$.

Solution: Simple reciprocal

Step 1: From the table: $\int dx/x = \ln|x| + C$

Step 2: For $x > 0$, $|x| = x$

$\ln x + C$.

Integrate √x

Find $\int \sqrt{x}\, dx$.

Solution: Integrate √x

Step 1: Write $\sqrt{x} = x^{1/2}$

Step 2: $\int x^{1/2}\,dx = x^{3/2} / (3/2) = (2/3)\,x^{3/2}$

$(2/3)\,x^{3/2} + C.$

Rough work / extra work

Integrate $1/x^2$

Find $\int (1/x^2)\, dx$.

Solution: Integrate $1/x^2$

Step 1: $1/x^2 = x^{(-2)}$

Step 2: $\int x^{(-2)}\,dx = x^{(-1)}/(-1) = -1/x$

$-1/x + C$.

Physics: integrating motion

Differentiation went downhill: $x \to v \to a$.

Integration goes uphill: $a \to v \to x$.

$$v = \int a\,dt + C_1 \quad \cdot \quad x = \int v\,dt + C_2$$

Initial conditions fix the constants C_1 and C_2.

From constant acceleration to velocity

If $a = 4\ \mathrm{m\ s^{-2}}$ (constant) and $v = 0$ at $t = 0$, find $v(t)$.

Solution: From constant acceleration to velocity

Step 1: $v = \int a\,dt = \int 4\,dt = 4t + C$

Step 2: At $t = 0$, $v = 0 \rightarrow 0 = 0 + C \rightarrow C = 0$

$v = 4t$ $(\mathrm{m\,s^{-1}})$.

Rough work / extra work

From velocity to position

If $v = 4t$ and $x = 0$ at $t = 0$, find $x(t)$.

Solution: From velocity to position

Step 1: $x = \int v\,dt = \int 4t\,dt = 2t^2 + C$

Step 2: At $t = 0$, $x = 0 \rightarrow C = 0$

$x = 2t^2$ (m).

Constant velocity displacement

If $v = 6\ \text{m s}^{-1}$ (constant) and $x = 2\ \text{m}$ at $t = 0$, find $x(t)$.

Solution: Constant velocity displacement

Step 1: $x = \int 6\,dt = 6t + C$

Step 2: $t = 0, x = 2 \rightarrow C = 2$

$x = 6t + 2$ (m).

Rough work / extra work

Area under a curve

A definite integral will later mean the signed area between the curve and the x-axis.

Area idea: $\int_a^b f(x)\,dx$ (limits a to b)

Tonight we stay with indefinite integrals only (no limits yet).

Do not mix indefinite (+C) with definite (number, no +C).

Homework style: power

Find $\int (3x^5 - x)\,dx$.

Solution: Homework style: power

Step 1: $\int 3x^5\,dx = 3 \cdot x^6/6 = x^6/2$

Step 2: $\int -x\,dx = -x^2/2$

$(1/2)x^6 - (1/2)x^2 + C.$

Homework style: trig mix

Find $\int (\sin x + \cos x)\,dx$.

Solution: Homework style: trig mix

Step 1: $\int \sin x\,dx = -\cos x$

Step 2: $\int \cos x\,dx = \sin x$

$-\cos x + \sin x + C$.

Rough work / extra work

Homework style: check

Differentiate $y = -\cos x + \sin x + 4$ and confirm it matches $\sin x + \cos x$.

Solution: Homework style: check

Step 1: $d(-\cos x)/dx = \sin x$

Step 2: $d(\sin x)/dx = \cos x$ · $d(4)/dx = 0$

$dy/dx = \sin x + \cos x$. Check successful.

Integration lock lines

Integration reverses differentiation

$$\int f(x)\,dx = F(x) + C$$

$$\int x^n\,dx = x^{n+1}/(n+1) + C \quad (n \neq -1)$$

Check every answer by differentiating

Practice

1. Memorise the standard integrals table from tonight.

2. Redo every ∫ question without looking at the solution.

3. For each answer, differentiate and recover the integrand.

4. Write one motion example: given a, find v and x with +C fixed by data.

Missing +C is the most common indefinite-integral mistake.